

Mobile Wireless Telecoms 2

Engineering Technology

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Mobile Wireless Telecoms 2 (A05667)

Short Title:	Mobile Wireless Telecoms 2	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Electronics	
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Credits	5	Level: Advanced

Description of Module / Aims

This module builds upon and continues on from the second year module entitled, Introductory Telecommunications Theory & the third year module entitled, Mobile Wireless Telecommunications-1. This module focuses primarily on advanced digital modulation techniques in the context of commercial cellular systems: 2G-Second Generation based modulation schemes (primarily GSM), 3G-Third Generation based modulation schemes (primarily Universal Mobile Telecommunications System (UMTS)), and 4G-Fourth Generation based modulation schemes (primarily Long Term Evolution (LTE) and Digital Video Broadcasting-Terrestrial (DVB-T)), and frames them in the context of the state of the art approach of Software Defined Radio (SDR). A practical program runs in parallel with lectures and tutorials to enhance and reinforce the student's understanding of the theory, using experience and technical information gleaned from the numerical modeling. This is realised by utilising off-the-shelf USB TV receivers, partnered with a standard PC, furnished with appropriate software suites (Python, C/C++, GnuRadio, etc (facilitating the PC with a Linux capability to access this hardware will be necessary)), as a laboratory workstation.

Programmes

WCOM-0001	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	4 / 7 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	4 / 7 / M
WCOM-0001	BSc (Hons) in Applied Electronics (WD_EELEC_B)	2 / 3 / M

Indicative Content

- Advanced Digital Modulation - 2nd Gen (2G): Partial Response Signalling (PRS) pulse shaping, spectral compression, narrowband constant envelope modulation. Architectural implementation of Gaussian MSK modulator & associated BER performance, matched filtering (equalization) demodulator requirement.
- Advanced Digital Modulation - 3rd Gen (3G): CDMA -- Direct Sequence & Frequency Hopping. Root Raised Cosine (RRC) wideband pulse shaping, rake receiver delay locked loop receiver architecture incorporating RRC matched filter. Spreading codes / selection criteria. Auto & cross correlation.
- Advanced Digital Modulation - Fourth Generation (4G): Orthogonal Frequency Division Multiplex (OFDM). MultiMode transmission & implications for wireless channel propagation, ISI. DVB-T, DRM, ADSL, LTE.
- Architecture / Infrastructure: SDR architecture and implementation. ADC architectures and principles for SDR. DVB-T USB module Radio Frequency front-end receiver (utilising Free-Software Foundation / GNU Radio software tools) for SDR. Toward MIMO architectures, Smart Antenna architectures.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discuss and differentiate between modulation and access schemes in terms of implementation and spectral efficiency.
2. Describe the theory for the need for advanced modulation schemes, the motivating factors and drivers for their employment in modern mobile wireless cellular telecommunications systems, and the technical implications of their implementation with current technology.
3. Express the physical layer modulation schemes associated with 2G, 3G & 4G cellular communications systems using advanced numerical modelling techniques.
4. Frame the digital modulation principles in the context of state of the art SDR implementation entities and verify this exercise through robust numerical models.
5. Construct numeric & circuit level simulations of telecommunications systems, such that a partnering physical layer apparatus can be constructed in a laboratory setting.
6. Write meaningful practical reports, commensurate with a prescribed format, that includes adequate presentation of numerical / circuit simulation and experimental results, relevant calculations and meta information, and draw relevant conclusions.

Learning and Teaching Methods

- Lectures
- Practicals
- Integrated Worked Examples
- Laboratory Work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,4
Continuous Assessment	40%	
<i>In-Class Assessment</i>	10%	1,2,4
<i>Practical</i>	30%	3,5,6

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Carlson, B.A., P.B. Crilly and J.C. Rutledge. _Communication Systems: An Introduction to Signals and Noise in Electrical Communication_. 4th ed.. : McGraw Hill, 2002.
- Feher, K. _Wireless Digital Communications_. Upper Saddle River, New Jersey: Prentice Hall, 1995.
- Glover, I. and P. Grant. _Digital Co__mmunications_. 2nd ed.. : Pearson Education, 2003.
- Lee, J.S. and L.E. Miller. _CDMA Systems Engineering Handbook_. 1st. Boston & London: Artech House Publishers, 1998.
- Sklar, B. _Digital Communications, Fundamentals and Applications_. 2nd ed.. : Prentice Hall, 2001.
- Steele, R. _Mobile Radio Communications_. 2nd Edition. New York, U.S.A.: Wiley-IEEE Press, 1999.

Requested Resources

- ENGINEERING LAB: Electronics